

POLYMARKET EXPECTED VALUE (EV) ENGINE

Quantitative Option Pricing and Volatility Arbitrage Framework

1. Executive Overview

The Polymarket Expected Value (EV) Engine is a high-frequency, autonomous quantitative trading framework built to trade short-term Bitcoin prediction contracts on decentralized platforms. In extremely short-term markets (e.g., 5-minute to 1-hour boundaries), standard price action analysis fails due to high noise. The EV Engine solves this by treating binary prediction contracts as European-style digital options, applying derivative option pricing theory to exploit discrepancies between real-world mathematical probabilities and the market share prices listed on Polymarket.

2. Option Pricing & Probability Formulation

At the core of the EV Engine is a volatility-scaled pricing model adapted from the Black-Scholes-Merton option pricing framework. The engine calculates the 'true mathematical probability' of Bitcoin closing above a specific strike price at a given expiration time.

2.1 Volatility Parameterization (Realized Volatility)

Standard volatility models fail in hyper-short-term trading. The EV Engine calculates live 5-minute volatility by parsing historical high, low, and close candles using the Average True Range (ATR) indicator over a 14-period window. This daily/hourly ATR is then scaled down to a per-second realized volatility:

$$\text{Volatility per Second} = (14\text{-period ATR} / \text{Spot Price}) / \sqrt{300}$$

2.2 Black-Scholes Probability Distance (d-score)

The standard option pricing model uses the cumulative normal distribution to map the spot price relative to the strike price over time. We calculate the probability distance (d1-like score) using spot price, strike price, time remaining in seconds (t), and per-second volatility (v):

$$d = \ln(\text{Spot Price} / \text{Strike Price}) / (\text{Volatility per Second} * \sqrt{\text{Time Remaining}})$$

2.3 True YES Option Probability

Using the mathematical Error Function (erf), we evaluate the cumulative distribution function (CDF) of the normal distribution. This yields the mathematically correct probability that Bitcoin will expire above the strike price:

$$\text{True YES Probability} = 0.5 * (1.0 + \text{erf}(d / \sqrt{2}))$$

The true NO probability is simply: True NO Probability = 1.0 - True YES Probability.

3. Expected Value (EV) Edge Arbitrage

Polymarket operates as a CLOB (Central Limit Order Book) where Yes and No shares trade between \$0.01 and \$0.99. If a YES share is priced at \$0.60, the market is pricing the event at a 60% probability. The EV Engine constantly compares its computed True Probability with the live market share prices to isolate Expected Value:

YES Expected Value (EV) = (True YES Probability * \$1.00) - Market YES Share Price
NO Expected Value (EV) = (True NO Probability * \$1.00) - Market NO Share Price

A positive EV indicates that the contract is mathematically underpriced relative to the asset's current price trajectory and volatility profile. The bot enforces a strict threshold where it only executes trades if the expected return edge is at least +\$0.02 USDC per share (representing a solid 2% minimum expected return edge).

4. Allora Network AI Synergy Integration

To augment the mathematical model, the EV Engine is fully integrated with the Allora Network API (Sepolia Topic 14 - BTC 5m price predictions). Allora acts as a machine-learning consensus layer:

- High-Probability Filters: If the Allora forecast predicted price deviates from spot by at least 0.05%, it represents a high confidence direction.
- Probability Boosting: When a high-confidence bullish AI signal is active, the engine overrides standard Black-Scholes limits, boosting True YES probability to 85% (and vice-versa for bearish signals).
- Win Rate Separation: AI-synergized trades are tagged as 'allora_...' in the SQLite database to track performance independently from pure math-driven trades.

5. Strict Risk Management and Settlement

- Fractional Kelly Risk: The engine dynamic scales trade sizing to risk exactly 1.0% of the live account balance per trade, calculated as: Risk Capital = Balance * 0.01.
- No Liquidation Risk: Since binary contracts do not carry leverage, the Stop Loss is set to 0.00, meaning positions cannot be liquidated early.
- Auto-Expiration Settlement: The backend thread tracks contract expiration in real-time, executing EIP-712 simulated payouts at \$1.00 or \$0.00 at the exact unix expiry second.

TIQ QUANTITATIVE PLATFORM

Autonomous Web3 Agent System Architecture | Powered by Oanda, Kraken & Allora Network